

Conformal Privacy Auditing: Calibrated Re-identification Attacks with Statistical Guarantees

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Abstract

Empirical identity leakage from released text is increasingly driven by attackers that combine large language models (LLMs) with auxiliary knowledge to link documents to individuals. Existing audits typically report success rates for specific attack pipelines but lack finite-sample statistical guarantees, while training-time protections such as differential privacy are difficult to translate into release-time decisions for individual natural-language documents. We introduce *Conformal Privacy Auditing* (CPA), a distribution-free calibration framework that provides a statistical certificate of re-identification risk for each released document against LLM-empowered adversaries. CPA outputs a *conformal ambiguity set* of candidate identities that is guaranteed to contain the true identity with user-chosen confidence under exchangeability, together with an interpretable leakage proxy derived from set size. CPA supports both logit-access and sampling-only attackers, enabling audits of open-source models and proprietary API models in a unified framework. Across multiple release benchmarks and attacker configurations, CPA achieves calibrated coverage and reveals sharp shifts in certified identifiability as auxiliary knowledge, LLM augmentation, and release mechanisms vary, providing a statistically grounded basis for reporting and comparing release-time linkage risk across attacker configurations, datasets, and release mechanisms alike.

1 Introduction

Privacy concerns have intensified alongside the rapid growth of large language models (LLMs) and their widespread applications (Xin et al., 2025). These applications are inherently data-intensive. In many domains, including healthcare and law, organizations face a strong need to release textual data, such as clinical notes (Lison et al., 2021), legal documents (Deuber et al., 2023), and online posts (Dou et al., 2024), to support research and product development. At the same time, they must ensure that releasing such data does not compromise the privacy of the individuals it describes.

A common practice for releasing sensitive textual data is redaction (Pilán et al., 2022), which replaces private information belonging to predefined categories with symbolic placeholders. However, as illustrated in Fig. 1, combinations of seemingly innocuous attributes, such as “37-year-old”, “female”, “nephrologist”, and “Brisbane”, can remain in redacted texts and collectively serve as linkable signals (Xin et al., 2025), enabling the re-identification of individuals who were intended to be anonymized. Recent studies (Manzanares-Salor et al., 2024; Staab et al., 2024) further show that the widespread use of LLMs has effectively turned them into practical privacy adversaries, capable of extracting salient cues from text and integrating auxiliary information to substantially amplify re-identification risks, even for carefully redacted text.

Existing approaches for *auditing* privacy risks in released textual data remain limited, particularly in the presence of LLMs with strong reasoning capabilities. Differential Privacy (DP) provides formal worst-case guarantees by injecting carefully calibrated noise into the data (Dwork, 2006). However, these guarantees are often overly conservative in practice, leading to significant utility degradation, while failing to capture the nuanced and context-dependent privacy risks observed in real-world deployments. In contrast, empirical privacy auditing methods quantify privacy leakage using attack-based metrics, such as re-identification success rates; such evaluations provide practical evidence of observed leakage but lack theoretical guarantees, leaving it unclear how well the measured risks generalize to unseen data or stronger attackers.

To address these challenges, we propose **Conformal Privacy Auditing** (CPA), a theoretically grounded *black-box* framework for auditing *re-identification* risks of individuals in released texts under a specified attacker model. CPA builds on the theory of Conformal Prediction (Vovk et al., 2005; Bates et al., 2021), a distribution- and model-agnostic framework for uncertainty quantification with formal statistical guarantees. As illustrated in Fig. 1, given a released document and attacker background knowledge, CPA employs a score function to identify a *certified ambiguity set* that covers the true target individual with a high probability, e.g., 90%, under the chosen attack model. The size of this ambiguity set serves as an interpretable measure of re-identification uncertainty—a large set (e.g., 10^6 candidates) indicates high uncertainty for the attacker to correctly identify

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the individual. Importantly, CPA provides *explicit statistical conditions* under which these guarantees hold, enabling principled interpretation and comparison of auditing outcomes across threat models.

The contributions of this work are three-fold.

- We formulate re-identification auditing of released texts as constructing *conformal ambiguity sets* that serve as privacy certificates, and develop a theoretical framework specifying the statistical conditions under which these certificates provably hold.
- We design nonconformity scores for LLM and retrieval-based attackers requiring only candidate log-probabilities or similarity scores, enabling fully *black-box* auditing of open-source models and proprietary API-based models without any internal or logit-level access.
- We demonstrate that CPA certifies privacy risks against a *declared family* of attackers, with formal statistical guarantees across diverse release benchmarks, attacker families, and adversarial settings.

2 Preliminaries: Conformal Prediction

Conformal prediction (CP) turns arbitrary model scores into *set-valued* predictions with distribution-free, finite-sample guarantees (Vovk et al., 2005). It requires no correctly specified probabilistic model; it relies only on *exchangeability*: the joint distribution of $(Z_1, \dots, Z_{n+1})$ with $Z_i = (X_i, Y_i)$ is invariant under permutations of indices (i.i.d. sampling is sufficient), so that a test example is statistically indistinguishable from the n examples of a held-out *calibration* sample and ranks of calibration scores form valid quantiles.

Concretely, fix any trained (possibly black-box) predictive system and a *nonconformity score* $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, where larger values indicate that label y is less compatible with input x . Given calibration scores $S_i = s(X_i, Y_i)$, split conformal sets the cutoff $\hat{q}_\alpha$ to the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest calibration score and returns the prediction set $C_\alpha(x) = \{y \in \mathcal{Y} : s(x, y) \leq \hat{q}_\alpha\}$. A standard rank argument yields the *finite-sample marginal coverage* guarantee

$$\Pr(Y_{n+1} \in C_\alpha(X_{n+1})) \geq 1 - \alpha, \quad (1)$$

which is distribution-free and holds for *any* score function s computed consistently on calibration and test examples (Vovk et al., 2005). In multiclass settings, adaptive prediction sets (APS) construct $s(x, y)$ from the model’s ranked probabilities so that sets shrink on easy examples and expand on hard ones while preserving marginal coverage (Romano et al., 2020).

Relation to threshold tuning on a hold-out set. Split CP resembles the familiar practice of tuning a classifier’s decision threshold on held-out data, but differs in what is guaranteed. Ad-hoc threshold tuning targets an empirical operating point (e.g., a precision/recall trade-off), makes no finite-sample statement about future data,

and inherits any miscalibration of the underlying score. Split CP instead specifies the exact order statistic whose use guarantees marginal coverage at level $1 - \alpha$ for the next exchangeable example—for any score function, without parametric or calibration assumptions. In our setting, this is what turns an arbitrary attacker score into a statistically calibrated ambiguity certificate with finite-sample validity guarantees.

3 Auditing Identity Linkage Risk

3.1 Running example: Release-time identity linkage auditing

Consider a healthcare provider releasing de-identified clinical case summaries (x) who must ensure that an attacker cannot link them to specific patients using auxiliary background knowledge ($\mathcal{R}$), such as demographics or public registries. The provider uses CPA to simulate a realistic linkage attacker that extracts clues with LLMs and ranks potential matches against a candidate profile database. For each document, the audit produces a *certified ambiguity set* $C_\alpha(x)$: a statistically guaranteed “short-list” of suspects containing the true patient with confidence $1 - \alpha$. A small set (e.g., $|C_\alpha(x)| = 3$) immediately signals that the attacker can narrow the identity to a tiny group, so the provider can withhold or further redact documents whose certified ambiguity falls below a minimum threshold k —directly mirroring government guidance that ties release decisions to numeric re-identification risk (e.g., the 0.09 threshold used by the Information and Privacy Commissioner of Ontario, 2016).

3.2 Task Formulation

Released documents and identities. We study *identity linkage risk* for released text. Let $X \in \mathcal{X}$ denote a released document (possibly anonymized or rewritten) and let $Y \in \mathcal{U}$ denote the true identity of the subject. Here $\mathcal{U}$ is a *reference population* (e.g., a candidate profile database). The goal of auditing is not to *protect* against an unbounded attacker, but to provide a *quantitative, finite-sample certificate* of linkage risk under a declared attacker capability and budget.

3.3 Threat Model

Every CPA certificate is issued *relative to a declared threat configuration*. Following standard practice in security research, we make each component of this configuration explicit below; Table 7 summarizes the knowledge ladder instantiated in our experiments, and Table 8 in Appendix D maps every experimental suite to its full configuration. A notation table consolidating the symbols used in this and the following sections is provided in Appendix C (Table 6) for reference.

Attacker goal. Given a released document x , the attacker attempts to link x to the true subject identity Y within the reference population $\mathcal{U}$.

4 Methodology: Conformal Privacy Auditing (CPA)

4.1 Nonconformity for Identity Linkage: APS Mass Score

We require a scalar score that measures how much probability mass must be accumulated to include the true identity under attacker A . Let $\pi_1(x, w), \pi_2(x, w), \dots$ be candidates sorted by decreasing $p_A(\cdot | x, w)$. We use an APS-style *mass score*, restricted to the candidate pool:

$$s(x, y; w) := \sum_{\substack{j: p_A(\pi_j | x, w) \\ \geq p_A(y | x, w)}} p_A(\pi_j | x, w) \in (0, 1], \quad (6)$$

where we abbreviate $\pi_j \equiv \pi_j(x, w)$. Smaller $s(x, y; w)$ indicates that the attacker concentrates probability on the truth (high linkage risk). Larger $s(x, y; w)$ indicates diffuse belief (lower linkage risk).

CPA is not tied to a single score. The calibration layer of CPA requires only an exchangeable scalar score; the APS mass score is one instantiation. In particular, the weights in (6) need not be calibrated probabilities: for retrieval attackers they are pseudo-posteriors obtained by softmax-normalizing similarity scores, and for ranking-only attackers a purely rank-based nonconformity score is equally valid. Our ablations (Section 7.1, Table 11) show that APS, rank-based, and raw-probability constructors all achieve valid coverage; we adopt APS as a robust default rather than claiming it is uniformly dominant.

Sampling-only interface. When logits are unavailable, we query the attacker m times with forced-choice decoding over $\mathcal{Y}(x, w)$ and form empirical frequencies $\hat{p}_m(\cdot | x, w)$. We then compute the plug-in score $\hat{s}_m(x, y; w)$ by replacing p_A with $\hat{p}_m$ in (6). This yields a fully black-box conformal pipeline applicable to proprietary LLM APIs that expose no logits.

4.2 Split Conformal Ambiguity Sets

Given calibration data $\{(X_i, Y_i, W_i)\}_{i=1}^n$ and a fixed knowledge level w (or conditioning on $W = w$), we compute scalar scores

$$T_i := \begin{cases} s(X_i, Y_i; w), & \text{probability/logits available,} \\ \hat{s}_m(X_i, Y_i; w), & \text{sampling-only.} \end{cases} \quad (7)$$

Let $T_{(1)} \leq \dots \leq T_{(n)}$ be the sorted scores and define the split-conformal threshold

$$k := \lceil (n+1)(1-\alpha) \rceil, \quad \hat{q}_\alpha := T_{(k)}. \quad (8)$$

For a new document (x, w) , we output the calibrated ambiguity set, i.e., the prediction set that contains the true identity with high confidence:

$$C_\alpha(x, w) := \{y \in \mathcal{Y}(x, w) : s(x, y; w) \leq \hat{q}_\alpha\}, \quad (9)$$

using s or $\hat{s}_m$ depending on access. Note that the inequality direction “ $\leq$ ” in (9) follows the *nonconformity* convention: top-ranked candidates have *smaller* cumulative mass scores, so the set retains all candidates that the attacker cannot statistically rule out. In particular, $\hat{q}_\alpha$ is a calibrated quantile of nonconformity scores, not a probability-of-linkage threshold. Section 5 gives the finite-sample coverage guarantee under exchangeability, together with its open-world extension.

4.3 Instantiations of the Attacker Pipeline

Our framework is attacker-agnostic: any pipeline that induces a distribution over candidates can be conformalized without any modification.

Retrieval-CPA (profile matching). We construct a profile text representation for each candidate profile (using a controlled schema) and define $p_A(\cdot | x, w)$ from normalized similarity scores (e.g., TF-IDF cosine similarity). Knowledge level w determines which profile fields and which query transformations are permitted (e.g., metadata-only vs. keyword profiles; with or without LLM clue extraction at query time).

Profile-CPA (LLM extraction $\rightarrow$ matching). An LLM attacker extracts attributes from x (e.g., occupation, location, age range), producing distributions over attribute values via sampling. These distributions induce probabilistic match scores against candidate profiles, yielding $p_A(\cdot | x, w)$ over identities. We then apply CPA exactly as in (8)–(9). This instantiation matches a realistic threat model in which LLMs infer quasi-identifiers and link to a candidate profile. The full procedure is listed in Algorithm 1 in Appendix A.1.

5 Privacy guarantee for CPA

This section formalizes the statistical guarantees provided by Conformal Privacy Auditing (CPA). Our results are distribution-free (no parametric assumptions on data) and apply to black-box attackers, provided an exchangeability assumption holds for the calibration and evaluation distributions under the declared attacker knowledge level w (Section 3.3).

5.1 Assumptions

Exchangeability (within a declared knowledge level).

Fix a knowledge level $w \in \mathcal{W}$ that specifies the attacker’s side information and tooling (Section 3.3). We assume that the sequence $\{(X_i, Y_i)\}_{i=1}^{n+1}$ used for calibration/evaluation is exchangeable under this fixed w (i.i.d. sampling is sufficient). Equivalently, in the most general form we assume $\{(X_i, Y_i, W_i)\}_{i=1}^{n+1}$ is exchangeable; when experiments are run stratified by category and knowledge level (as in our suites), $W_i \equiv w$ is constant and exchangeability reduces to exchangeability of (X_i, Y_i) within that stratum.

Attacker pipeline and randomness. For each example i , the attacker pipeline (retrieval or LLM-based) may involve internal randomness—e.g., sampling-only

forced-choice decoding, stochastic clue extraction, or stochastic rewriting. We denote this randomness by U_i and assume: (i) U_i are i.i.d. across examples, and (ii) U_i are independent of the data (X_i, Y_i, W_i) . This reflects operational practice: each audit query uses an independent random seed / sampling stream, and the auditor does not adversarially choose randomness based on the content of any individual example.

Candidate-pool miss probability. In open-world linkage, the true identity may be absent from the candidate pool $\mathcal{Y}(X, w)$. We measure this using the miss probability

$$\Pr(Y \notin \mathcal{Y}(X, w)) \leq \rho(w), \quad (10)$$

reported empirically as `candidate_miss_rate`. Our strongest guarantees apply conditional on $Y \in \mathcal{Y}(X, w)$ and degrade gracefully with the miss probability $\rho(w)$, as shown in Corollary 5.3.

5.2 CPA Validity for a Fixed Attacker Configuration

Recall the APS-style nonconformity score (restricted to $\mathcal{Y}(x, w)$):

$$s(x, y; w) = \sum_{\substack{y': p_A(y' | x, w) \\ \geq p_A(y | x, w)}} p_A(y' | x, w) \in (0, 1].$$

In the sampling-only interface, p_A is replaced by empirical frequencies $\hat{p}_m$, yielding $\hat{s}_m(x, y; w)$. CPA uses split conformal calibration: compute scores on a calibration set, take $\hat{q}_\alpha$ as the $\lceil (n+1)(1-\alpha) \rceil$ -th order statistic, and return $C_\alpha(x, w) = \{y : s(x, y; w) \leq \hat{q}_\alpha\}$.

Lemma 5.1 (Score exchangeability under sampling-only access). *Let $\{(X_i, Y_i, W_i)\}_{i=1}^{n+1}$ be exchangeable and let U_i be i.i.d. independent randomness used by the attacker pipeline to compute $\hat{s}_m(X_i, Y_i; W_i)$. Define scalar scores*

$$T_i := \hat{s}_m(X_i, Y_i; W_i) = f(X_i, Y_i, W_i, U_i)$$

for a deterministic measurable function f (deterministic given the example and the sampled outputs). Then $\{T_i\}_{i=1}^{n+1}$ is an exchangeable sequence of scalar scores.

Proof sketch. Augment each example to (X_i, Y_i, W_i, U_i) . Exchangeability of (X_i, Y_i, W_i) and i.i.d. independence of U_i implies the augmented sequence is exchangeable. Since T_i is a deterministic function of the augmented example, exchangeability is preserved under deterministic measurable mappings of the entire augmented example sequence. $\square$

Theorem 5.2 (Finite-sample CPA validity for identity linkage). *Fix a knowledge level w and attacker pipeline A . Assume: (i) exchangeability holds for calibration and test examples under w , (ii) $Y \in \mathcal{Y}(X, w)$ almost surely (closed-world or $\rho(w) = 0$), and (iii) sampling randomness (if used) satisfies Lemma 5.1. Let $\hat{q}_\alpha$ be the*

split-conformal threshold computed from n calibration scores at level α , and let $C_\alpha(\cdot, w)$ be defined by the rule $s(x, y; w) \leq \hat{q}_\alpha$ (or $\hat{s}_m$ in the sampling-only case). Then for a new test example (X_{n+1}, Y_{n+1}) ,

$$\Pr(Y_{n+1} \in C_\alpha(X_{n+1}, w)) \geq 1 - \alpha. \quad (11)$$

Proof sketch. Let T_i denote the scalar scores used for conformalization. By exchangeability (Lemma 5.1 if needed), the rank of T_{n+1} among $\{T_1, \dots, T_n, T_{n+1}\}$ is uniform. With $\hat{q}_\alpha$ chosen as the k -th order statistic for $k = \lceil (n+1)(1-\alpha) \rceil$, we obtain $\Pr(T_{n+1} \leq \hat{q}_\alpha) \geq 1 - \alpha$. Membership $Y \in C_\alpha(X, w)$ is equivalent to $T \leq \hat{q}_\alpha$. $\square$

Corollary 5.3 (Open-world validity with candidate-miss probability). *Fix w and suppose $\Pr(Y \in \mathcal{Y}(X, w)) \geq 1 - \rho(w)$. If conditional on the event $E = \{Y \in \mathcal{Y}(X, w)\}$ the CPA set satisfies $\Pr(Y \in C_\alpha(X, w) | E) \geq 1 - \alpha$, then*

$$\begin{aligned} \Pr(Y \in C_\alpha(X, w)) &\geq (1 - \rho(w))(1 - \alpha) \\ &\geq 1 - (\rho(w) + \alpha). \end{aligned} \quad (12)$$

Proof. By the law of total probability, $\Pr(Y \in C_\alpha) = \Pr(E) \Pr(Y \in C_\alpha | E)$, and apply the two bounds. $\square$

Theorem 5.2 states that CPA outputs a *valid ambiguity set* containing the truth with probability at least $1 - \alpha$ under the declared attacker knowledge level w . The ambiguity proxy $1/|C_\alpha|$ is therefore a calibrated *certificate of attacker ambiguity*, not a claim about universal privacy against unbounded attackers.

6 Experiments

Goal. We evaluate *conformal privacy auditing* as a reporting layer for document release. Given a released text x and an attacker-side *candidate profile database* $\mathcal{R}$ that encodes auxiliary knowledge, our auditor outputs a *certified ambiguity set* $C_\alpha(x) \subseteq \mathcal{R}$. Intuitively, $|C_\alpha(x)|$ quantifies how many plausible identities remain after the attack pipeline is applied, while the finite-sample conformal guarantee ensures calibrated uncertainty under exchangeability.

6.1 Datasets and Release Mechanisms

We use datasets where each released document is associated with a ground-truth identity/profile, enabling identity linkage evaluation. For each dataset we construct (i) released documents x (e.g., anonymized text) and (ii) a candidate profile database $\mathcal{R}$ of candidate profiles built from *non-released* sources (e.g., original text, metadata, or historical documents), reflecting attacker side information. Because certified ambiguity is only meaningful relative to the candidate pool, we describe the per-dataset candidate-pool construction explicitly below; Table 8 in Appendix D maps every experimental suite to its full threat configuration.

TextWash. TextWash (Kleinberg et al., 2022) provides paired orig and anon person descriptions. We treat anon as the released document and build the candidate profiles from orig. The benchmark includes three categories (famous, semi-famous, fictional), which we interpret primarily as differing *availability of external context*: external web knowledge is abundant for famous individuals, limited for semi-famous, and absent for fictional entities. *Candidate pool*: one profile per identity in the same category, constructed from the orig description (with identity tokens removed); linkage is evaluated within-category, so the pool size equals the size of the category (≈ 400 candidate profiles).

TAB (text anonymization benchmark). TAB (Pilán et al., 2022) consists of text records where anonymization may preserve substantial lexical content. We use TAB as an easy leakage regime to verify that the auditor reports near-singleton ambiguity sets when linkage is straightforward. *Candidate pool*: all 1,268 case records, each represented by a profile built from metadata only (K1), extracted keywords (K2), or case text (K3), with no verbatim copying of the released document into the profiles.

WikiBio. WikiBio (Stranisci et al., 2023) provides biography-style text with structured fields. We use WikiBio to control *profile completeness* by varying which fields are exposed to the attacker. *Candidate pool*: 1,000 candidate profiles built from infobox-style fields or extracted keywords, audited against 500 released biographies drawn from the prepared slice.

Blog Authorship Corpus. We use the Blog Authorship corpus (Schler et al., 2006) as a large-scale linkage setting. To keep experiments tractable, we subsample to a fixed number of authors and released posts. *Candidate pool*: one profile per author, built by concatenating W held-out posts ($W \in \{1, 3, 10\}$) that are disjoint from the released posts, so that W directly controls the strength of attacker side information in this suite.

Candidate-pool truncation and miss rate. For expensive attacker pipelines we optionally restrict conformal scoring to the top- K retrieved candidates—an *engineering choice* that can violate the closed-world condition of Theorem 5.2 if the true identity is dropped. We therefore report the candidate-miss rate with both *conditional* coverage (given the truth is retained) and *unconditional* coverage, and quantify the effect of the truncation level K in the stress tests of Section 7.1.

6.2 Attack Pipelines and Knowledge Levels

Our threat model is a *declared attacker pipeline* that maps a released document to a ranked list or distribution over candidates, given side information such as locations, gender, and occupations (Section 3.3). We study two complementary families of attackers.

Retrieval attacker. We build a text index over profiles. Given a released document x , the attacker forms a query $q(x)$ and retrieves a ranked list of candidates.

We instantiate a *knowledge ladder* by varying the information available in profiles and the query construction: (i) a minimal schema subset (weak), (ii) a richer profile description including auxiliary facts (medium), and (iii) query augmentation using LLM-extracted clues from x (strong). For TextWash famous/semi-famous, we additionally consider a web-assisted attacker that can enrich the query $q(x)$ with web search results.

LLM attribute/clue attacker (sampling-based). We also evaluate an LLM that extracts attributes or salient clues from x via repeated sampling. These predictions are matched against the profiles to produce candidate scores. We treat the number of LLM samples m as a controllable attacker resource and study its impact on the resulting audit outcomes.

Models. We use two open-source instruction-following LLMs as local attackers (Llama-3.1-8B-Instruct (Grattafiori et al., 2024) and Qwen3-8B-Instruct (Yang et al., 2025)), and one stronger cloud model (GPT-5). Local models are used for clue extraction and attribute prediction in the sampling-based attacker. The retrieval attacker does not require an LLM unless explicitly augmented.

Metrics. We report both non-conformal ranking metrics and conformal auditing metrics. *Non-conformal baselines.* We report Top-1/Top- k accuracy, mean reciprocal rank (MRR), and mean rank of the true identity in the retrieved list. *Conformal privacy auditing.* For each $\alpha \in \{0.01, 0.05, 0.1\}$ we report empirical coverage $\frac{1}{N} \sum_{i=1}^N \mathbf{1}\{y_i \in C_\alpha(x_i)\}$, the distribution of set sizes $|C_\alpha(x)|$ (median and tail percentiles), and a leakage proxy $1/\max\{1, |C_\alpha(x)|\}$. When the candidate pool may not contain the truth, we also report the empirical candidate-miss rate for each suite.

6.3 Protocol and Reproducibility

For each dataset and attacker configuration we use a split conformal protocol with disjoint calibration and test partitions. Because the conformal threshold $\hat{q}_\alpha$ is an order statistic of the calibration scores, it depends on the calibration size; we therefore treat the split sizes as part of the declared configuration rather than as fixed parameters, state them explicitly wherever attackers are compared (e.g., $n_{\text{cal}}=n_{\text{test}}=100$ per TextWash category), and ablate the effect of n_{cal} in Section 7.1. When the attack pipeline includes sampling (LLM attribute/clue extraction), sampling randomness is generated independently per example, with a declared per-document sampling budget m . We calibrate a separate conformal threshold per dataset and knowledge level, enabling statistically grounded comparison across threat models without re-tuning attack heuristics. Attacker-to-attacker comparisons (e.g., direct retrieval vs. LLM-assisted clues) are made only on *matched* calibration/test budgets, since certified set sizes are not comparable across different split sizes. Whenever the release distribution or the attacker pipeline changes—including prompt, model,

Attacker setting	Top-1	Cov.	Med. $ C $	Mean $ C $	Mean $1/ C $
TAB-Direct					
K1 meta / direct	0.052	0.929	1268	1179.5	9.57e-04
K2 keywords / direct	0.973	0.973	1	1	1.0000
K3 case-text / direct	0.994	0.994	1	1	1.0000
K2 keywords / LLM-clues	0.715	0.970	20	20.1	0.0499
TAB-Direct (Quasi-ID)					
K1 meta / direct	0.003	0.932	1268	1173.6	0.0017
K2 keywords / direct	0.872	0.953	4	3.7	0.2717
K3 case-text / direct	0.964	0.964	1	1	1.0000
K2 keywords / LLM-clues	0.615	0.945	27	27.1	0.0369

Table 1: TAB linkage auditing ($\alpha = 0.05$). Smaller ambiguity sets indicate higher certified identifiability. *direct* uses the released document as the retrieval query; *LLM-clues* uses an LLM to extract a short clue string from the released document before retrieval. *meta/keywords/case-text* describe which fields are available in the candidate profile database. All suites use split conformal calibration with disjoint calibration/test partitions; the full declared threat configuration per suite is summarized in Table 8 in Appendix D.

or web-snapshot changes—we treat the result as a *new* attacker configuration and recalibrate before issuing certificates.

7 Experimental Results

CPA yields comparable, threat-model-specific certificates. Across tables and datasets, the empirical coverage is generally close to the target $1 - \alpha$ (e.g., 0.95 when $\alpha = 0.05$), supporting that CPA produces a statistically meaningful certificate that is comparable across attacker configurations rather than a single, uncalibrated, attack-specific success rate reported in isolation from the declared threat model.

Release mechanism $\times$ auxiliary knowledge jointly determine certified identifiability. Table 1 shows a sharp transition in certified identifiability on TAB as the attacker’s auxiliary knowledge increases. With *weak side information* (K1: metadata-only profiles), CPA returns near-full ambiguity sets (median $|C_\alpha|$ close to the full candidate pool), certifying substantial residual uncertainty even if the attacker occasionally ranks the true identity highly. When the attacker is granted *richer profile signals* (K2: keyword profiles; K3: case-text profiles), the ambiguity sets collapse toward singletons, certifying that the *same released documents* become highly linkable under stronger auxiliary knowledge. The quasi-identifier release variant exhibits the expected mitigation effect: compared to direct release, CPA sets expand (e.g., median $|C_\alpha| \approx 4$ under K2 keywords), indicating reduced—but still non-trivial—linkability. Overall, TAB illustrates the central message of CPA: privacy risk is not an intrinsic property of the released text alone, but depends critically on the attacker’s declared side information and tooling.

The miscoverage level α is a practical audit “dial” that trades certificate strength for ambiguity. Figure 2 (Appendix E) visualizes the conformal trade-off between statistical confidence and the size of the cer-

tified ambiguity set. On TAB (direct retrieval under the quasi-identifier release), tightening the guarantee (smaller α) yields larger ambiguity sets: at $\alpha=0.01$, empirical coverage is 0.998 and the median set size is 19 (low leakage proxy $\mathbb{E}[1/|C_\alpha|] \approx 0.052$). Relaxing the guarantee shrinks sets and increases the leakage proxy: at $\alpha=0.05$, the median set size drops to 4 ($\mathbb{E}[1/|C_\alpha|] \approx 0.233$), and at $\alpha=0.1$ the median reaches 1 ($\mathbb{E}[1/|C_\alpha|] \approx 0.825$). Once α becomes sufficiently large, CPA enters a near-singleton regime: ambiguity sets cannot shrink below size 1, so the conformal certificate effectively coincides with the attacker’s point prediction and empirical coverage approaches the attacker’s Top-1 accuracy. This provides a clean interpretation for auditors: *smaller α certifies residual ambiguity more conservatively; larger α yields smaller certified sets at correspondingly weaker confidence.*

Blog Authorship shows the same qualitative “dial” behavior: with only coarse demographic-style profile fields (“W_demo”), calibration saturates ($\hat{q}_\alpha = 1$) and CPA returns near-full candidate sets across α , whereas with text-enriched profiles (“W_text”, $W=10$ posts/author) the median $|C_\alpha|$ drops from 9 ($\alpha=0.01$) to 7 ($\alpha=0.05$) to 5 ($\alpha=0.1$), with coverage decreasing accordingly as the guarantee weakens.

LLM-assisted clue extraction can amplify linkage—but the effect is model-dependent and visible only with set-valued certificates. Table 2 shows that augmenting retrieval with LLM-generated clues can either reduce ambiguity sets (higher certified identifiability) or fail to help, depending on the attacker model. For WikiBio, Llama-based clue extraction increases Top-1 accuracy and shrinks the median ambiguity set relative to direct retrieval, whereas Qwen-based clues behave closer to the direct baseline. This highlights a key methodological point for LLM-era privacy auditing: different plausible attacker models can induce meaningfully different linkage risks, and CPA offers a principled way to compare them under a common calibrated out-

Table 2: Retrieval-CPA on WikiBio (500 released bios; 1k candidates) under direct query vs LLM-assisted clue extraction ($\alpha = 0.05$), with split conformal calibration over disjoint calibration/test partitions of the released bios; the full declared threat configuration is summarized in Table 8 in Appendix D.

Attacker setting	Top-1	Coverage	Median $ C_\alpha $	Mean $ C_\alpha $	$\mathbb{E}[1/ C_\alpha]$	$\hat{q}_\alpha$
Direct query	0.362	0.956	264	254.98	0.00466	0.277
LLM clues (Llama-3.1-8B)	0.420	0.950	189	186.87	0.00562	0.2
LLM clues (Qwen3-8B)	0.380	0.958	265	255.69	0.00465	0.277

put. A similar, but more nuanced, pattern appears on TextWash under matched evaluation budgets (Table 5): LLM clues reduce certified ambiguity only where external context exists, making shifts in *certified residual uncertainty* visible where fixed Top- k summaries would show little or no measurable change.

Scale and cohort effects: narrowing the candidate pool can dominate the certificate. When the candidate pool is large, certified ambiguity sets remain large even with enriched side information (Table 4), indicating substantial residual uncertainty at the chosen confidence level. In contrast, in “motivated intruder” scenarios where the pool is implicitly narrowed (organizational, demographic, or geographic constraints), CPA sets can collapse dramatically, certifying much higher linkability. Realistic auditing should therefore report results under multiple candidate-pool assumptions, not only under a single global pool.

TextWash Retrieval-CPA under matched budgets. Table 5 reports conformal privacy auditing results on TextWash at $\alpha = 0.05$, with direct retrieval and LLM-clue attackers evaluated on the *same* capped split ($n_{\text{cal}} = n_{\text{test}} = 100$ per category), so that certified set sizes are directly comparable across attackers. Direct retrieval yields large certified ambiguity sets (median $|C_\alpha|$ of 120–169.5 depending on category), indicating that even when Top-1 accuracy is moderate, the attacker still cannot be *certifiably* confident about a small identity set. The effect of LLM-assisted clue extraction is *category- and model-dependent* rather than uniform: for famous subjects, where external context exists, Llama clues shrink the certified median from 120 to 90 (Top-1 0.470 $\rightarrow$ 0.510), whereas for fictional and semi-famous subjects—where little external knowledge is available—clue extraction leaves the certified sets essentially unchanged relative to direct retrieval. We therefore do not claim that LLM clues uniformly dominate direct retrieval; rather, CPA makes visible *where* LLM augmentation genuinely increases certified identifiability. Empirical coverage remains close to the nominal target (≈ 0.95) across all matched runs.

7.1 Robustness under Stressed Assumptions

We now stress each assumption behind the CPA guarantee empirically, making the boundary of the certificate visible rather than only theoretical (full tables and per-suite breakdowns in Appendix E).

Exchangeability and attacker/release drift. We calibrate under one configuration and test under another (Table 9). Matched configurations meet the $1 - \alpha = 0.95$ target (TAB direct $\rightarrow$ direct 0.965; Blog W10 $\rightarrow$ W10 0.940), while shifted configurations fall below it (TAB direct $\rightarrow$ quasi-ID 0.923; Blog W10 $\rightarrow$ W1 0.835). A CPA certificate is thus calibrated for a declared threat configuration and does *not* automatically transfer under attacker or release drift—recalibration is the operational protocol. More robust variants (e.g., Mondrian or weighted conformal prediction) are possible when the drift model is explicit, but do not provide distribution-free transfer under arbitrary prompt, model, or snapshot drift, which remains an open problem for conformal methods.

Candidate-pool miss and top- K truncation. Sweeping the truncation level K (Table 10) separates conformal miscoverage from candidate-pool failure. For TextWash fiction with direct retrieval at $K = 50$, the candidate-miss rate is 0.160: coverage *conditional* on candidate inclusion remains 1.000, while unconditional coverage drops to 0.840; at $K = 200$ the miss rate vanishes and unconditional coverage recovers to 0.970 (WikiBio+Llama behaves analogously). Closed-world validity (Theorem 5.2) applies conditional on candidate inclusion, and the observed degradation matches the multiplicative bound in Corollary 5.3.

Alternative score constructors at matched coverage. Replacing the APS mass score with rank-based, raw-probability, or calibrated top- K constructors over cached LLM-clue queries (Table 11) shows that multiple constructors achieve valid coverage: on WikiBio+Llama at $\alpha = 0.05$, APS attains coverage 0.950 (median $|C_\alpha| = 189$), rank-based 0.950 (median 179), and raw probability 0.960 (median 212). CPA’s calibration layer is therefore not tied to a single score; APS is a robust default rather than uniformly dominant, and rank-based scores extend CPA to attackers that expose nothing but a ranked list of candidate identities.

Calibration-set size. Varying $n_{\text{cal}} \in \{50, 100, 150\}$ on matched TextWash direct retrieval at $\alpha = 0.05$ (Table 12) yields average coverage 0.962/0.957/0.927 with average median set sizes 151.2/148.2/140.3: coverage stays near target while sets tighten with more calibration data. Since $\hat{q}_\alpha$ is an order statistic of the calibration scores, certified set sizes are comparable only

at matched splits; we state the split sizes explicitly for all matched comparisons and ablations.

Sampling budget and split stability. Increasing the sampling-only budget m from 1 to 10 (Llama-3.1-8B-Instruct clue extraction on TextWash famous; Table 13) keeps coverage above target (0.970–0.980) while the median certified set size drops from 120 to 78: the budget affects *efficiency*, not validity, once recalibrated. Across three random calibration/test splits, coverage is stable at 0.951 ± 0.037 (median $|C_\alpha|$ 143.3 ± 33.6). Prompt, model, or decoding changes are treated as attacker-configuration drift requiring recalibration.

Operational interpretation. The key advantage of CPA over fixed Top- k reporting is that it turns a ranked list into a *certificate*: for a chosen α , the auditor can interpret $|C_\alpha(x)|$ as the (calibrated) residual uncertainty remaining after the attack. This supports concrete release policies such as “withhold or further redact documents with median $|C_\alpha| < k$ under attacker configuration w ,” and encourages tail-focused auditing of the smallest, highest-risk certified sets first.

8 Related Work

Differential privacy for training and fine-tuning language models. Differential privacy (DP) provides a formal stability-based guarantee for randomized algorithms (Dwork et al., 2006), with DP-SGD (Abadi et al., 2016) as the standard training approach, adapted to NLP fine-tuning and generation (Yu et al., 2022; Li et al., 2022). These works enforce a mechanism-level guarantee; in contrast, we audit *task-level* re-identification risk under a declared threat model, with finite-sample guarantees derived from split conformal calibration rather than from mechanism design or calibrated noise injection.

Unintended memorization and data extraction attacks. Language models can memorize and regurgitate training data or sensitive spans, enabling extraction-style attacks (Carlini et al., 2019, 2021, 2023), which motivates attack-based evaluation protocols such as membership inference and canary tests (Shokri et al., 2017; Yeom et al., 2018). Our approach is compatible with these attack definitions: any attack score can serve as the underlying nonconformity signal.

Privacy auditing and post hoc evaluation. A parallel literature audits trained models to estimate or falsify privacy claims, including auditing of DP pipelines (Jagielski et al., 2020; Steinke et al., 2023). Hu et al. (2025) suggest that recovering a single scalar “privacy number” may be brittle across settings. Our method instead outputs a *set-valued certificate* with a coverage guarantee under a declared threat family.

Text anonymization benchmarks and re-identification. The Text Anonymization Benchmark (TAB) provides a corpus and evaluation framework

for reducing disclosure risk in released text (Pilán et al., 2022), and recent work argues anonymization should be assessed via de-anonymization experiments with realistic adversaries (Deuber et al., 2023). We complement these efforts with a distribution-free *risk-control layer*: given a threat model and attacker pipeline, the reported ambiguity sets have guaranteed finite-sample coverage on held-out data.

Statistical disclosure control and record linkage. Release-time disclosure risk has a long history in statistical disclosure control (SDC) and record linkage: probabilistic record linkage scores matches against a candidate list (Fellegi and Sunter, 1969); disclosure-risk estimation quantifies per-record re-identification probability (Lambert, 1993; Reiter, 2005; Hundepool et al., 2012); and k -anonymity bounds risk via a minimum cohort of indistinguishable records (Sweeney, 2002). CPA shares this candidate-list view, and its certified ambiguity set can be read as a calibrated, per-document analogue of a k -anonymity cohort under a *modern, declared* attacker. Unlike classical SDC, which targets structured microdata and model-based population-frequency assumptions, CPA operates on unstructured text attacked by retrieval- and LLM-based pipelines and replaces model-based risk estimates with a distribution-free, finite-sample set-valued coverage guarantee. CPA does not replace classical SDC analysis; it wraps a declared modern attacker pipeline with a statistical certificate that classical disclosure-risk metrics by themselves do not provide for released text.

Conformal prediction and risk control. Conformal prediction provides finite-sample, distribution-free uncertainty quantification under exchangeability (Vovk et al., 2005), with adaptive prediction sets controlling coverage while adjusting set size to difficulty (Romano et al., 2020) and conformal *risk control* extending to generic risks (Bates et al., 2021). CPA adapts this principle to privacy: the risk is re-identification under a declared threat model and the output is certified attacker ambiguity instead of a point estimate.

9 Conclusion

We studied *release-time* privacy risk for released text against black-box attackers that combine LLMs with auxiliary knowledge for identity linkage. Our framework, *Conformal Privacy Auditing* (CPA), wraps arbitrary attacker pipelines into a *finite-sample privacy certificate*, valid in both logit-access and sampling-only settings and thus applicable to open-source and proprietary API models alike. Empirically, CPA delivers calibrated coverage across benchmarks and shows that privacy risk is heterogeneous—a release that looks safe under limited side information can become highly identifiable under richer registries or stronger reasoning—so evaluation should favor calibrated, threat-model-specific certificates over fragile point estimates.

Limitations

CPA is an *auditing* framework, not a cryptographic privacy definition. Its guarantees hold under the declared threat model: a finite candidate pool and an attacker pipeline family specified by the auditor, together with an exchangeability assumption between calibration and evaluation examples within each configuration. In particular, CPA does not assume the auditor can enumerate all future attackers, and it provides *no* universal protection against unknown or unanticipated attacks—the history of linkage attacks shows that even expert practitioners underestimate attacker ingenuity. Each certificate is valid only for the declared candidate pool, attacker pipeline, knowledge level, and calibration/test distribution; we therefore recommend running CPA across multiple plausible attacker families and recalibrating whenever attacker tooling, external knowledge, or release mechanisms change. Our drift stress tests (Section 7.1) quantify how coverage degrades when this protocol is not followed. If the candidate pool omits the true identity (open-world miss) or if deployment data drift violates exchangeability, coverage can degrade; we therefore report candidate-miss rates together with conditional and unconditional coverage, and recommend recalibration when the release distribution changes (e.g., after rewriting). More robust calibration variants (e.g., Mondrian or weighted conformal prediction under covariate shift) are possible when the drift model is explicit, but they require additional assumptions and do not provide distribution-free transfer under arbitrary prompt or model drift. Finally, this work focuses empirically on identity linkage; the same calibration layer applies to any attack that yields a scalar score or ranked candidate set, and extending CPA to membership inference is a natural direction for future work. These limitations are not unique to CPA but reflect the fundamental challenge of giving operationally meaningful, release-time guarantees for natural language under evolving adversaries.

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A Appendix

A.1 CPA Algorithm

The full CPA procedure is shown in Algorithm 1.

Algorithm 1 Split Conformal Privacy Auditing (CPA)

Require: Calibration set $\mathcal{D}_{\text{cal}} = \{(X_i, Y_i)\}_{i=1}^n$ under fixed knowledge level w ; candidate-pool generator $\mathcal{Y}(\cdot, w)$; attacker pipeline A ; miscoverage level $\alpha \in (0, 1)$; sampling budget m (used only in sampling-only access).

Ensure: Conformal threshold $\hat{q}_\alpha$ and ambiguity-set function $C_\alpha(\cdot, w)$.

1: **Score function (APS mass).** For any (x, y) define

$$s(x, y; w) = \sum_{\substack{y' \in \mathcal{Y}(x, w): \\ p_A(y'|x, w) \geq p_A(y|x, w)}} p_A(y' | x, w).$$

2: **Calibration scores.**

3: **for** $i = 1$ to n **do**

4: Construct candidate pool $\mathcal{Y}_i \leftarrow \mathcal{Y}(X_i, w)$.

5: Obtain a distribution $\tilde{p}_i(\cdot) \approx p_A(\cdot | X_i, w)$ over $\mathcal{Y}_i$:

6: **if** logits/logprobs available **then**

7: $\tilde{p}_i \leftarrow p_A(\cdot | X_i, w)$ restricted/normalized to $\mathcal{Y}_i$.

8: **else** {sampling-only access}

9: Draw $\tilde{Y}_i^{(1)}, \dots, \tilde{Y}_i^{(m)} \sim A(\cdot | X_i, w)$ over $\mathcal{Y}_i$.

10: Set $\tilde{p}_i(y) \leftarrow \frac{1}{m} \sum_{j=1}^m \mathbf{1}\{\tilde{Y}_i^{(j)} = y\}$ for $y \in \mathcal{Y}_i$.

11: **end if**

12: Compute calibration score $T_i \leftarrow \sum_{y' \in \mathcal{Y}_i: \tilde{p}_i(y') \geq \tilde{p}_i(Y_i)} \tilde{p}_i(y')$.

13: **end for**

14: Sort $T_{(1)} \leq \dots \leq T_{(n)}$ and set

$$k \leftarrow \lceil (n+1)(1-\alpha) \rceil, \quad \hat{q}_\alpha \leftarrow T_{(k)}.$$

15: **Define ambiguity set for a new document** (x, w) .

16: Construct $\mathcal{Y}(x, w)$ and obtain $\tilde{p}(\cdot) \approx p_A(\cdot | x, w)$ as above.

17: Compute plug-in scores $\tilde{s}(y)$ for all $y \in \mathcal{Y}(x, w)$ as in Step 1 (with $\tilde{p}$ in place of p_A), and output

$$C_\alpha(x, w) \leftarrow \{y \in \mathcal{Y}(x, w) : \tilde{s}(y) \leq \hat{q}_\alpha\}.$$

18: **leakage proxy:** $L(x, w) \leftarrow 1 / \max\{1, |C_\alpha(x, w)|\}.$

B Dataset Statistics

Table 3 summarizes the datasets used in our privacy-auditing experiments, together with their split statistics.

The TAB benchmark contains 1014 training documents, 127 development documents, and 127 test documents. These records contain long judicial narratives,

which is reflected in the substantially larger average document length relative to the other datasets.

The TextWash benchmark contains 1202 original and 1202 released descriptions, with category counts of 401 famous, 400 semi-famous, and 401 fiction/other examples. Average document length is approximately 130 words before release and 128 words after release.

The Blog authorship corpus is stored as 19320 author-level XML files. Parsing those XML files yields an average of 9.74 posts per author and a median of 6 posts per author. This corpus is substantially larger in author count than the other datasets and is used to study how CPA behaves when the attacker registry is constructed from varying numbers of posts per author.

The WikiBio experiments use the prepared audit slice emitted by the dataset-preparation script. It contains 1000 released biographies with 1000 unique identities, and the matching profile database contains 1000 profile rows. The average original and released biography lengths are 102.7 and 101.9 words respectively, reflecting the lighter editing typical of the prepared WikiBio audit slice used in our experiments.

B.1 Implementation Details

Retrieval attacker (non-LLM and LLM-assisted).

In Retrieval-CPA, the attacker maps a released document x to a probability distribution over a candidate pool by: (i) building candidate profile texts for each identity in the auxiliary database, (ii) embedding them with a bag-of-words TF-IDF vectorizer, and (iii) ranking candidates by cosine similarity to a *query string* derived from x . We convert similarity scores to a pseudo-posterior distribution via a temperature-scaled softmax (parameter `-softmax_temperature`).

Direct query (`-query_mode direct`) uses the released text itself as the query string. *LLM clues* (`-query_mode direct_plus_clues`) uses an LLM to extract a short list of missing/high-value clues from the released text (optionally augmented by web search), and appends them to the query string. This is implemented in `adversary/clue_extract.py` and parsed via `linkage/clues.py` within `privacy_audit_cp`. In both cases, we can optionally restrict to a top- K candidate pool (`-top_k_pool`) to reduce runtime while preserving a well-defined candidate set for the split conformal calibration performed at audit time.

Profile attacker (LLM attribute extraction + candidate filtering).

In Profile-CPA, the LLM adversary does not see (and is not prompted with) the full candidate list. Instead, it predicts attributes from the released text (e.g., gender, age-range, occupation, location, or dataset-specific fields) by sampling m times per attribute. The sampled attribute values induce empirical distributions $\hat{p}_m(\cdot | x)$. For each attribute, we conformalize these distributions to produce a certified set of plausible attribute values; the final identity ambiguity set is formed by filtering the auxiliary profile database to candidates consistent with *all* certified attribute sets.

Dataset	Unit	Total	Split / subset	Avg. words
TAB (ECHR)	documents	1,268	train/dev/test = 1014/127/127	train 1343.4; dev 874.6; test 843.4
TextWash	biographies	1,202 orig + 1,202 anon	famous 401, semifamous 400, fiction/other 401	orig 130.5; anon 127.7
Blog Authorship	authors	19,320	balanced male/female author roster	9.74 posts/author
WikiBio	biographies	1,000	one category in the prepared audit slice	orig 102.7; anon 101.9

Table 3: Dataset statistics for the release-audit experiments. For TAB we report split sizes from the benchmark JSON files; for TextWash we report file counts from the original and anonymized person-description trees; for Blog we report author-level XML statistics; and for WikiBio we report the prepared audit slice used by the canonical experiment runner in all reported WikiBio suites and ablations.

Conformal calibration and outputs. Both Retrieval-CPA and Profile-CPA use split conformal calibration: we compute a scalar nonconformity score for the true identity (or true attribute value) on a calibration split and take an order statistic threshold $\hat{q}_\alpha$. At test time we output an ambiguity set $C_\alpha(x)$ consisting of candidates whose score is below $\hat{q}_\alpha$. Per run we report: (i) empirical coverage, (ii) distribution of $|C_\alpha(x)|$ (median/mean/min/max), (iii) *leakage proxy* $\mathbb{E}[1/\max\{1, |C_\alpha(x)|\}]$, and (iv) diagnostic rates such as empty-set and candidate-miss (when applicable). For reproducibility and debugging, the runner scripts write jsonl files for both calibration and test splits containing the query string, top- K ranked candidates, conformal scores, and the final ambiguity set for every audited document in both conformal splits.

Randomness and the U_i variables. Whenever an attacker uses sampling (e.g., m LLM generations for clue extraction or attribute extraction), the randomness is represented abstractly as a per-example seed/noise variable U_i . In code, U_i corresponds to the randomness consumed by stochastic decoding (and any parsing post-processing). Conditional on (X_i, W_i) and the fixed prompt/decoding settings, the resulting score T_i is a deterministic function of (X_i, Y_i, W_i, U_i) , which is exactly the condition needed by the exchangeability-based conformal validity argument in Section 5.

Default hyperparameters. Unless otherwise stated, we use: TF-IDF with `max_features=50,000`, `ngram_max=1`, and dataset-dependent `min_df`; softmax temperature in $[0.5, 2.0]$; LLM decoding with `temperature=0.8`, `top_p=0.95`; and $m \in \{5, 10, 30\}$ samples depending on the ablation. We use fixed random seeds for dataset splits and sampling (whenever seeding is supported by the inference backend).

Blog Authorship: attacker side-information via held-out posts. Table 4 reports Retrieval-CPA on Blog Authorship under a controlled increase in attacker side information. Each released document is a post whose true identity is its author. To model auxiliary knowledge, we build a *candidate database* of author profiles by concatenating W held-out posts per author ($W \in \{1, 3, 10\}$), and run a retrieval attacker that ranks authors by text similarity between the released post and each author profile. We then apply split conformal calibration at $\alpha = 0.05$ to convert the attacker ranking into a certified ambiguity

set $C_\alpha(x)$. As W increases, the attacker becomes stronger: Top-1 accuracy rises ($0.087 \rightarrow 0.209$) and the conformal threshold $\hat{q}_\alpha$ decreases ($0.948 \rightarrow 0.763$), yielding smaller certified ambiguity sets (median $|C_\alpha|$ drops from 222 to 176). Importantly, empirical coverage remains near the nominal $1 - \alpha \approx 0.95$, indicating that the certificates remain valid while exposing how additional auxiliary text reduces *certified* uncertainty. Despite this trend, the certified sets remain large in absolute terms, suggesting substantial residual ambiguity in a large candidate pool even when the attacker has multiple posts per author at its disposal.

C Notation

Table 6 consolidates the symbols used in Sections 3–5.

D Threat-Model Configurations per Experimental Suite

Table 7 summarizes the knowledge ladder used to instantiate attacker side information, and Table 8 maps each experimental suite to its declared threat configuration (Section 3.3): candidate-pool construction, query mode, LLM/web augmentation, and closed- vs. open-world assumption. The conformal split sizes and sampling budgets of the matched and ablation suites are stated in the corresponding table captions (Tables 5, 12, and 13). Each CPA certificate is valid for exactly one row of this table; changing any entry defines a new attacker configuration that requires recalibration.

E Robustness and Ablation Studies

This section reports the stress tests and ablations summarized in Section 7.1, together with the α -sweep visualization for the TAB quasi-identifier suite (Figure 2).

E.1 Exchangeability and Attacker/Release Drift

Table 9 calibrates CPA under one configuration and evaluates it under another. For Blog, $W=k$ denotes an attacker registry built from k known posts per author, so W10 is a stronger author-profile registry than W1. Matched configurations meet the $1 - \alpha = 0.95$ target, while shifted configurations fall below target, confirming that certificates do not transfer across attacker or release drift and must be recalibrated.

Table 4: Retrieval-CPA on Blog Authorship under increasing attacker side information (number of posts per author; $\alpha = 0.05$), with split conformal calibration over disjoint calibration/test partitions of the released posts; the full declared threat configuration is summarized in Table 8 in Appendix D.

Attacker setting	Top-1	Coverage	Median $ C_\alpha $	Mean $ C_\alpha $	$\mathbb{E}[1/ C_\alpha]$	$\hat{q}_\alpha$
W=1 posts/author	0.087	0.946	222	200.50	0.00566	0.948
W=3 posts/author	0.143	0.956	210	191.82	0.00604	0.903
W=10 posts/author	0.209	0.946	176	147.14	0.00677	0.763

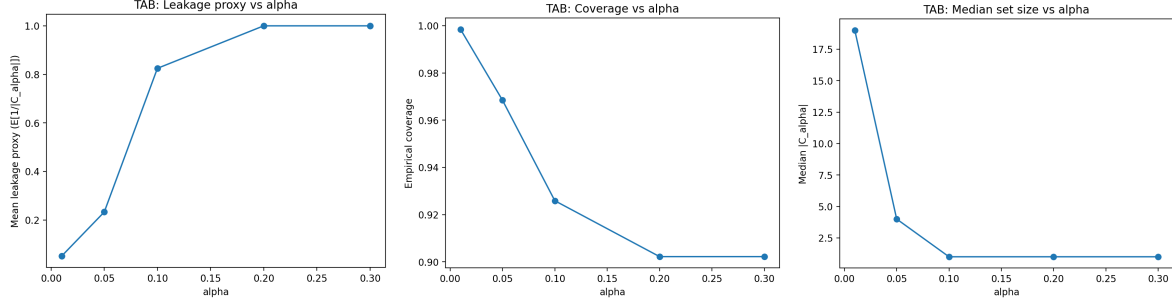


Figure 2: Effect of the miscoverage level α on TAB (direct retrieval, quasi-identifier release): mean leakage proxy $\mathbb{E}[1/|C_\alpha|]$ (left), empirical coverage (center), and median certified set size $|C_\alpha|$ (right). Smaller α certifies residual ambiguity more conservatively; larger α yields smaller sets with weaker statistical protection.

Table 5: **TextWash Retrieval-CPA under matched budgets.** Retrieval-CPA on TextWash under direct retrieval vs. LLM-assisted clue extraction ($\alpha = 0.05$). All attackers are evaluated on the *same* capped split with $n_{\text{cal}} = n_{\text{test}} = 100$ per category, so certified set sizes are directly comparable across attackers; LLM rows use cached clue outputs throughout.

Attacker	Top-1	Cov.	Med. $ C_\alpha $	Mean $ C_\alpha $
fiction				
Direct query	0.500	0.970	169.5	172.44
GPT-5 clues	0.500	0.970	169.5	172.44
Llama clues	0.600	0.970	171.0	174.26
semifamous				
Direct query	0.380	0.910	155.0	167.12
GPT-5 clues	0.380	0.910	155.0	167.12
famous				
Direct query	0.470	0.990	120.0	119.88
Llama clues	0.510	0.960	90.0	89.70

Smaller $|C_\alpha|$ implies a more identifying attack (less ambiguity). Coverage should be near $1 - \alpha = 0.95$ under exchangeability, up to finite-sample fluctuation.

E.2 Candidate-Pool Miss and Top- K Truncation

Table 10 sweeps the top- K truncation level, reporting the candidate-miss rate, coverage conditional on the true identity being retained in the pool, and unconditional coverage. Conditional coverage stays at or above target whenever the true identity is retained, while unconditional coverage degrades with the miss rate, exactly as predicted by Corollary 5.3. This separates conformal miscoverage from candidate-pool construction failure:

top- K truncation is an engineering choice, and closed-world validity applies conditional on candidate inclusion in the truncated candidate pool.

E.3 Alternative Score Constructors

Table 11 compares nonconformity-score constructors at matched coverage over cached LLM-clue queries (no new API calls): the APS mass score, a rank-based score, and the raw attacker probability. Multiple constructors achieve valid coverage, confirming that the CPA calibration layer is not tied to a single score; APS is a robust default rather than uniformly dominant, and the rank-based variant extends CPA to attackers that return only ranked candidate lists without scores.

E.4 Calibration-Set Size

Table 12 varies the calibration size n_{cal} on matched TextWash direct retrieval ($\alpha = 0.05$, $n_{\text{test}} = 100$ per category). Coverage remains near target while the certified sets tighten with more calibration data; per-category behavior (e.g., famous coverage $0.980 \rightarrow 0.940$ with median $124 \rightarrow 97$ as n_{cal} grows from 50 to 150, while fiction remains at $0.980/0.970/0.980$) shows the expected finite-sample variability of the order-statistic threshold. Because $\hat{q}_\alpha$ depends on n_{cal} through the split-conformal order statistic, the split sizes are part of the declared threat configuration and are stated with every matched comparison and ablation.

E.5 Sampling Budget and Split Stability

Table 13 reports a local Llama-3.1-8B-Instruct clue-extraction ablation on TextWash famous (direct-plus-clues, $n_{\text{cal}} = n_{\text{test}} = 100$, candidate-miss rate 0) over the per-document sampling budget m . Increasing m

Symbol	Meaning
x, X	Released document (possibly anonymized/rewritten)
y, Y	True identity of the document subject
$\mathcal{U}$	Reference population of identities
w, W	Attacker knowledge level / threat configuration (side information and tooling)
$\mathcal{Y}(x, w)$	Finite candidate pool available to the attacker
K	Top- K truncation level of the candidate pool (engineering choice)
$p_A(\cdot \mid x, w)$	Attacker’s distribution (or normalized ranking) over $\mathcal{Y}(x, w)$
m	Per-document sampling budget in the sampling-only interface
$\hat{p}_m$	Empirical attacker distribution from m forced-choice samples
$s(x, y; w), \hat{s}_m$	APS mass nonconformity score (exact / plug-in)
T_i	Scalar calibration score of example i
α	Target miscoverage level
$\hat{q}_\alpha$	Split-conformal threshold ($\lceil (n + 1)(1 - \alpha) \rceil$ -th order statistic)
$n_{\text{cal}}, n_{\text{test}}$	Calibration / test split sizes
$C_\alpha(x, w)$	Certified ambiguity set (primary certificate)
$L(x, w)$	Inverse ambiguity proxy $1/\max\{1, C_\alpha(x, w) \}$ (not a re-identification probability)
$\rho(w)$	Candidate-miss probability $\Pr(Y \notin \mathcal{Y}(X, w))$
U_i	Attacker-internal randomness (sampling, decoding) for example i

Table 6: Notation used throughout the paper.

from 1 to 10 preserves above-target coverage while reducing the median certified set size from 120 to 78: the sampling budget affects efficiency after recalibration, not validity. Across three random calibration/test splits of TextWash direct retrieval (nine category-by-seed rows), coverage is 0.951 ± 0.037 with median $|C_\alpha|$ 143.3 ± 33.6 (mean 146.5 ± 35.2), indicating stability to the split randomness. We treat prompt, model, or decoding changes as attacker-configuration drift requiring recalibration (Table 9); we do not compare or interpret certificates across such configuration changes.

Table 7: Knowledge ladder (side information W) used to instantiate attacker pipelines in our experiments.

Level	Attacker access / capability
K1 (weak)	Profiles contain only coarse structured fields (e.g., occupation bucket, coarse location, age range). Retrieval query uses released text directly.
K2 (medium)	Profiles add additional non-identifying evidence facts (e.g., coarsened events/keywords). Retrieval uses released text; no LLM assistance.
K3 (strong)	LLM-assisted clue extraction from released text produces a query expansion (optional), and/or profile is augmented (RAG/web search).

Suite	Candidate pool (size)	Query mode	LLM/web aug.	World
TAB K1-K3	case records (1268)	direct	none	closed
TAB K2/K3 + clues	case records (1268)	direct+clues	LLM clues	closed
WikiBio direct	profile rows (1000)	direct	none	closed
WikiBio + clues	top- $K=200$ of 1000	direct+clues	LLM clues	top- K truncated
Blog $W \in \{1, 3, 10\}$	author registry	direct	none	closed
TextWash direct (matched)	category (≈ 400)	direct	none	closed
TextWash + clues (matched)	category (≈ 400)	direct+clues	LLM clues	closed
TextWash m -ablation	category (≈ 400)	direct+clues	LLM clues (local HF)	closed

Table 8: Declared threat configuration for each experimental suite. “direct” uses the released text as the retrieval query; “direct+clues” appends LLM-extracted clues. The matched TextWash suites use $n_{\text{cal}}=n_{\text{test}}=100$ per category (Table 5), and the m -ablation sweeps the per-document sampling budget $m \in \{1, 3, 5, 10\}$ (Table 13).

Cal. $\rightarrow$ test	Cov.	Med. $ C_\alpha $	Mean $ C_\alpha $
TAB direct $\rightarrow$ direct	0.965	1	1.000
TAB direct $\rightarrow$ quasi-ID	0.923	1	1.000
TAB quasi-ID $\rightarrow$ quasi-ID	0.965	3	2.658
Blog W10 $\rightarrow$ W10	0.940	176	175.691
Blog W10 $\rightarrow$ W1	0.835	176	175.224

Table 9: Exchangeability stress tests ($\alpha = 0.05$). Matched calibration/test configurations achieve target coverage; shifted configurations under-cover, so recalibration is the operational protocol under suspected attacker-side or release-side drift.

Suite	K	Miss	Cond.	Uncond.	Med.
TW fiction direct	50	0.160	1.000	0.840	50
TW fiction direct	100	0.080	1.000	0.920	100
TW fiction direct	200	0.000	0.970	0.970	169.5
TW famous direct	50	0.110	1.000	0.890	50
TW famous direct	100	0.030	1.000	0.970	100
WB + Llama clues	50	0.118	1.000	0.882	50
WB + Llama clues	200	0.044	0.994	0.950	182

Table 10: Candidate-pool / top- K truncation stress tests ($\alpha = 0.05$). TW = TextWash, WB = WikiBio. “Miss” is the empirical candidate-miss rate; “Cond.”/“Uncond.” are coverage conditional on the true identity being retained in the truncated pool and unconditional coverage; “Med.” is the median certified set size $|C_\alpha|$ over the evaluated test split.

Attacker	Score	Cov.	Med.	Mean
WikiBio + Llama	APS	0.950	189	186.870
WikiBio + Llama	rank	0.950	179	179.000
WikiBio + Llama	prob.	0.960	212	209.434
Blog W10 + Llama	APS	0.950	8	8.341
Blog W10 + Llama	rank	0.962	9	9.000
TAB K3 + clues	APS	0.967	31	30.803
TAB K3 + clues	prob.	0.951	19	24.465

Table 11: Score-constructor ablation for cached LLM-clue attackers ($\alpha = 0.05$). “Med.”/“Mean” refer to $|C_\alpha|$. All constructors achieve valid coverage; efficiency (certified set size) varies moderately across constructors and across the different attacker pipelines.

n_{cal}	Avg. coverage	Avg. median $ C_\alpha $
50	0.962	151.2
100	0.957	148.2
150	0.927	140.3

Table 12: Calibration-set-size ablation on matched TextWash direct retrieval ($\alpha = 0.05$, $n_{\text{test}} = 100$), averaged over the three TextWash categories.

m	Top-1	Cov.	Med. $ C_\alpha $	Mean $ C_\alpha $
1	0.490	0.980	120	120.020
3	0.510	0.970	83	82.590
5	0.510	0.970	82	82.110
10	0.490	0.970	78	78.060

Table 13: Sampling-budget ablation with Llama-3.1-8B-Instruct on TextWash famous ($\alpha = 0.05$, $n_{\text{cal}} = n_{\text{test}} = 100$, candidate-miss rate 0). Coverage stays above target while larger budgets yield tighter certified sets at the same nominal coverage level.